INCH-POUND

MIL-C-63513A (AR)
AMENDMENT 5
27 July 1991
SUPERSEDING
AMENDMENT 4
23 January 1989

MILITARY SPECIFICATION

CARTRIDGE, 105MM, TP-T, M490A1, LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-63513A (AR), dated 5 September 1985 and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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* 3.2: Delete drawing number "9343009" and substitute "9343009-1 or 9343009-2".

PAGE 6

* First Article Inspection: Delete drawing number "9343009" and substitute "9343009-1 or 9343009-2".

PAGE 7

* First Article Inspection: Delete drawing number "9343009" and substitute "9343009-1 or 9343009-2".

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4.4.2.1:

* Delete drawing number "9343011" and substitute "9343011 or 12934834".

* Next Higher Assembly: Delete "9343010" and substitute "9343010-1 or 9343010-2".

* Minor 202: Delete "Evidence of poor workmanship, including liner film torn or damaged" and substitute "Evidence of poor workmanship, including torn or damaged liner film, skirt or string".

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4.4.2.2:

* Delete drawing number "9343010" and substitute "9343010-1 or 9343010-2".

* Next Higher Assembly: Delete "9343009" and substitute "9343009-1 or 9343009-2".

* Add Major 106: "106 Evidence of contamination in the inside diameter of the cartridge case 100% 3.8 Visual".

Change "Length of liner incorrect (Flush to 1/4 inch above mouth of case)" to "Length of liner incorrect) Flush to 1/2 inch above mouth of case)."

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4.4.2.3 Critical Defect: Add: "2 Plug missing 100% 3.2 Visual".

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4.4.2.4:

* Delete drawing number "9343009" and substitute "9343009-1 or 9343009-2".

Delete: "201 Evidence of poor workmanship 1.65% 3.8 Visual" and substitute: "201 Evidence of poor workmanship 0.65% 3.8 Visual".

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4.4.2.5:

* Delete drawing number "9343009" and substitute "9343009-1 or 9343009-2".

* Major 102: Delete Examination or Test column in its entirety and substitute: "Liner flaps not folded over propellant and cemented or secured with draw strings".

* Major 103: After examination or test item, add "(Note 1)".

* Add Major 104: "104 Liner skirts not tightened over propellant with proper knot 100% 3.2 Visual".

* Add Note 1: "1. Only one inspection is applicable depending on the method of closure utilized."

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* 4.4.2.6: Delete drawing number "9343009" and substitute "9343009-1 or 9343009-2".

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4.4.2.7:

* Delete drawing number "9343009" and substitute "9343009-1 or 9343009-2".

Delete: "104 Gap between cartridge case and obturator, max. 0.40% 3.2 Gage" and substitute: "104 Gap between cartridge case and obturator, max. 100% 3.2 Gage".

Add Major 105: "105 Forward bourrelet diameter, max. (Note 4) 100% 3.1 Ring Gage".

Add Major 106: "106 Runout of cartridge assembly NA 3.2 Gage/4.5.5.1".

Add Note 4: "4. To be performed immediately prior to packaging."

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* 4.5.1.2: Add new paragraph 4.5.1.2 as follows:

"4.5.1.2 Contamination in the inside diameter Prior to assembly of liner to cartridge case, solvent wipe (Acetone, Spec O-A-5) inside diameter where the adhesive/liner touches the cartridge case. Solvent and wiper (lint free) shall be visually clean."

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4.5.5 Delete the entire paragraph and substitute the following:

"4.5.5 Profile and alignment check. The profile and alignment of each crimped cartridge shall be verified by a gage corresponding to drawing 8828098, which is to be horizontal ($0^{\circ} \pm 2.5^{\circ}$). The rear face of the cartridge shall be positioned within one inch of the gage's push arm and be inserted with a peak force not to exceed 60 pounds (27.2 kg) and a peak velocity of 2 ft/sec (61 cm/sec). No manual assistance (i.e., rotation, agitation, etc.) is permitted during inspection. Any cartridge which fails to be inserted or seated with the required force and velocity in any orientation and in accordance with the following sampling plan shall be rejected:

a. Inspect (50) cartridges in 4 angular orientations approximately 90° apart.

b. After (50) consecutive cartridges successfully chambergage, switch to chambergaging 1 out of 100 in 4 angular orientations and all others in 1 (one) random orientation.

c. Any failure will result in returning to step (a.).

When applicable, rejected cartridges which are reworked or repaired in accordance with Government approved procedures shall be reinspected in 4 angular orientations approximately 90° apart. Failure of cartridge to be inserted or seated in any orientation shall be cause for rejection of the cartridge. The profile and alignment gage shall not be utilized as an alignment tool."

Add the following paragraph:

"4.5.5.1 Runout inspection. Any cartridge that exceeds the runout dimension per drawing 9343009 shall be chambergaged in 4 angular orientations approximately 90 degrees apart (see 4.5.5). This feature shall be controlled using Statistical Process Controls in accordance with the Contractor's government approval plan."

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